

Full Stack Development Tasks & Instructions — CodeAlpha

Full Stack Development — Internship Overview

This internship program provides comprehensive experience in both frontend and backend web development. CodeAlpha is a leading software development company focused on building end-to-end scalable web applications. The internship empowers students to work with modern technologies like HTML, CSS, JavaScript, React (or Angular/Vue) for frontend, and Node.js, Express.js, or Django for backend. Interns will gain hands-on experience in RESTful API development, database integration, authentication, and deployment. With expert mentorship and real-world projects, interns will develop the skills to build complete full-stack applications from scratch.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete source code to **GitHub** in a repository named:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn with GitHub repo link.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 3 or 2 out of the 4 tasks** listed below (from your domain).
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Full Stack Development Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Simple E-commerce Store

Build a **basic e-commerce site** with product listings.

Frontend: **HTML, CSS, JavaScript**.

Backend: Use **Django** (Python) or **Express.js** (Node.js).

Add features like:

- **Shopping cart**
- **Product details page**
- **Order processing**

- **User registration/login**

Database for **storing products, users, and orders**.

✓ TASK 2: Social Media Platform

Create a **mini social media app**.

Features to implement:

- **User profiles**
- **Posts & comments**
- **Like/follow system**

Frontend: **HTML, CSS, JavaScript**

Backend: **Django or Express.js**

Database for **users, posts, comments, followers**.

✓ TASK 3: Project Management Tool

Build a **collaborative tool** similar to Trello or Asana.

Users should be able to:

- **Create group projects**
- **Assign tasks**
- **Comment and communicate** within tasks

Full stack with **auth system, project boards, task cards**.

Backend to manage **users, projects, tasks, comments**.

Bonus: Add **notifications and real-time updates** using WebSockets.

✓ TASK 4: Real-Time Communication App

A **video conferencing + collaboration tool**.

Core features:

- **Video calling** (multi-user)
- **Screen sharing**
- **File sharing**
- **Whiteboard** for drawing/writing
- **Data encryption**
- **User authentication**

Tools/Libraries:

- WebRTC for **real-time media**
- Socket.io or Firebase for **real-time communication**
- Secure backend for **auth & data handling**

Gain experience across **frontend, backend, media streaming, and security**.



Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**.

Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.

Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.

Contact Information

- Website: www.codealpha.tech
- WhatsApp: [+91 9336576683](https://wa.me/919336576683)
- Email: services@codealpha.tech , services.codealpha@gmail.com